how natural log and e cancels out each other

The **natural logarithm** $\ln$ and the **exponential with base** e cancel each other because they are **inverse functions**. Think of them as operations that undo each other.

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1. Key Property

$$\ln(e^x) = x$$

and

$$e^{\ln(x)} = x \quad (x > 0)$$

This means applying one after the other brings you back to the original number.

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2. Why This Happens

- The natural log **asks the question**:
  "To what power must e be raised to get this number?"

Example:

$$\ln(e^5)$$

means:

> What power of e gives e^5?

Answer: **5**

So:

$$\ln(e^5) = 5$$

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3. Reverse Case

$$e^{\ln(7)}$$

means:

- $\ln(7)$ is the power you raise e to get 7.

So:

$$e^{\ln(7)} = 7$$

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4. Simple Analogy

Think of **square and square root**:

$$\sqrt{x^2} = x$$

They cancel because they undo each other.

Similarly:

- e^x → exponential
- $\ln(x)$ → natural log

They are **inverse operations**.

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5. Quick Examples

$$\ln(e^3) = 3$$

$$e^{\ln(10)} = 10$$

$$\ln(e^{2x}) = 2x$$

✅ Rule to remember

- If **ln is outside**, remove e and keep the exponent.
- If e **is outside**, remove ln and keep the inside.

If you want, I can also show a **visual graph explanation (why they are inverses)** which makes this even clearer. 📈

explain through proof like coomon log or log with any base and e cancels eachother or not also give more deep explanation how they cancel out like log x =1 if x=10 because log base and x if same then the power of x will be the answer

Let's go **step-by-step with a mathematical proof**, starting from the definition of logarithms and then showing **why e and $\ln$ cancel**.

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1. Definition of Logarithm

A logarithm is defined as:

$$\log_b(x) = y \quad \text{if and only if} \quad b^y = x$$

Meaning:

> The logarithm tells you **the power to which the base must be raised to get the number**.

Example:

$$\log_{10}(100) = 2$$

because

$$10^2 = 100$$

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2. Special Case: Natural Log

The **natural logarithm** is simply log with base e:

$$\ln(x) = \log_e(x)$$

So by definition:

$$\ln(x) = y \quad \text{if} \quad e^y = x$$

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3. Proof that $\ln(e^x) = x$

Start with:

$$y = \ln(e^x)$$

Using the definition of natural log:

$$e^y = e^x$$

Now since exponential functions with the same base are **one-to-one**, their exponents must be equal:

$$y = x$$

Therefore:

$$\ln(e^x) = x$$

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4. Proof that $e^{\ln(x)} = x$

Let:

$$y = \ln(x)$$

From the definition of natural log:

$$e^y = x$$

Substitute y:

$$e^{\ln(x)} = x$$

So again they **undo each other**.

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5. Why Any Log Cancels With Its Base

The same idea works for **any base**.

Definition:

$$\log_b(x) = y$$

means

$$b^y = x$$

Proof

Let

$$y = \log_b(b^x)$$

By definition:

$$b^y = b^x$$

Since bases are the same:

$$y = x$$

Therefore:

$$\log_b(b^x) = x$$

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6. Reverse Property

$$b^{\log_b(x)} = x$$

Proof:

Let

$$y = \log_b(x)$$

Then by definition:

$$b^y = x$$

Substitute y:

$$b^{\log_b(x)} = x$$

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7. Example with Common Log

Common log is base 10.

$$\log(10^4) = 4$$

because

$$10^4 = 10^4$$

Reverse:

$$10^{\log(7)} = 7$$

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8. Why They "Cancel"

They cancel because they are **inverse functions**.

A function and its inverse always satisfy:

$$f^{-1}(f(x)) = x$$

and

$$f(f^{-1}(x)) = x$$

Here:

- $f(x) = e^x$
- $f^{-1}(x) = \ln(x)$

So:

$$\ln(e^x) = x$$

$$e^{\ln(x)} = x$$

9. Deep Intuition

Logarithm asks:

> **"What power of the base produces this number?"**

Example:

$$\ln(e^5)$$

means

> What power of e gives e^5?

Answer:

$$5$$

So the log simply **extracts the exponent**.

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10. General Rule

For any base $b > 0, b \neq 1$:

$$\log_b(b^x) = x$$

and

$$b^{\log_b(x)} = x$$

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✅ **Key Insight**

Logarithm and exponent cancel **because they reverse each other's operation**.

Exponentiation:

$$b^x$$

Logarithm:

$$\log_b(x)$$

Logarithm **undoes exponentiation**.

If you want, I can also show the **change-of-base proof using calculus** that explains **why $\ln(x)$ is the most natural logarithm and why e appears everywhere in mathematics**. 📘